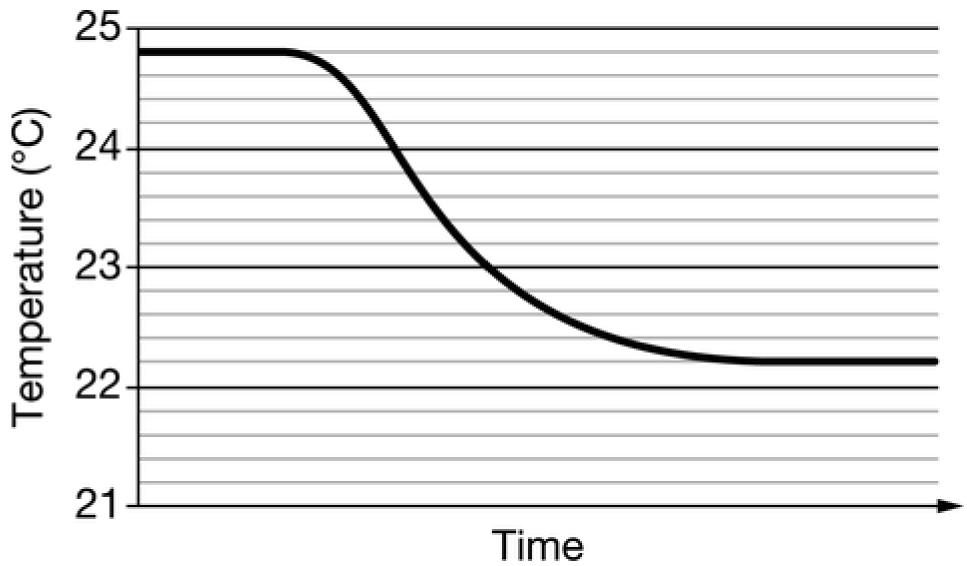
Unit 6 Progress Check: FRQ

1. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.



A student added a sample of $NH_4NO_3(s)$ to water and measured the temperature over time. The data are shown in the graph above.

(a) What is the value of ΔT that the student should use to calculate q ?

Please respond on separate paper, following directions from your teacher.

(b) According to the graph, is the dissolution of $NH_4NO_3(s)$ endothermic or exothermic? Explain.

Please respond on separate paper, following directions from your teacher.

(c) Are the strengths of the interactions between the particles in the solute and between the particles in the solvent before the solute and solvent are combined greater than, less than, or equal to the strengths of the interactions between solute particles and solvent particles after dissolution? Explain.

Please respond on separate paper, following directions from your teacher.

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Part (a)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response indicates that ΔT is $2.6°\text{C}$ or $-2.6°\text{C}$.

Part (b)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response indicates that the dissolution is endothermic because the temperature decreased and the energy of a system decreases when an endothermic process occurs.

Part (c)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response indicates indicates one of the following criteria:

- ☐ In an endothermic reaction, the strengths of the interactions between the solute particles and between the solvent particles are greater than the strength of the interactions between solute and solvent particles.
- ☐ The strengths of the interactions between the solute particles and between the solvent particles are greater than the strength of the interactions between solute and solvent particles" (and the response further explains that the change in energy is equal to the sum of the energies required to separate solute

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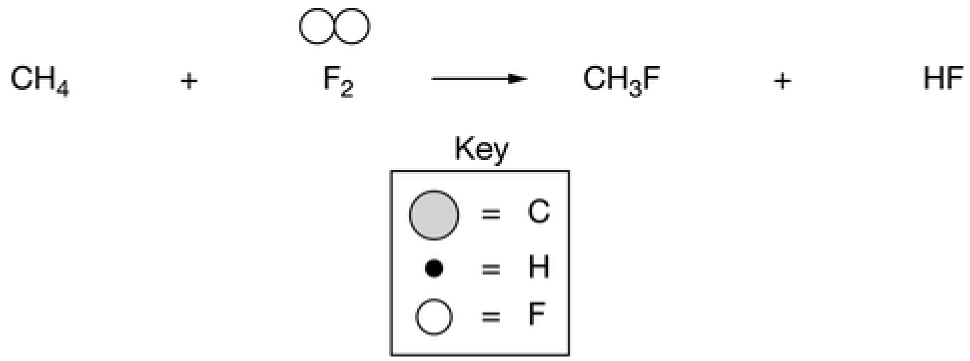
particles from one another and solvent particles from one another minus the energy released when attractions between solute particles and solvent particles form).

2. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

$CH_4(g)$ gas reacts with $F_2(g)$ to produce $CH_3F(g)$ and $HF(g)$.

(a) A particulate representation of $F_2(g)$ is shown above its formula in the equation below. Using the key provided, draw particulate representations of $CH_4(g)$ and the two product molecules of the reaction above their formulas in the equation.



Please respond on separate paper, following directions from your teacher.

(b) Use the bond enthalpies in the table below to calculate a numerical estimate of ΔH for the reaction.

Bond	Bond Enthalpy (kJ/mol)
$C-H$	413
$C-F$	485
$H-F$	567
$F-F$	155

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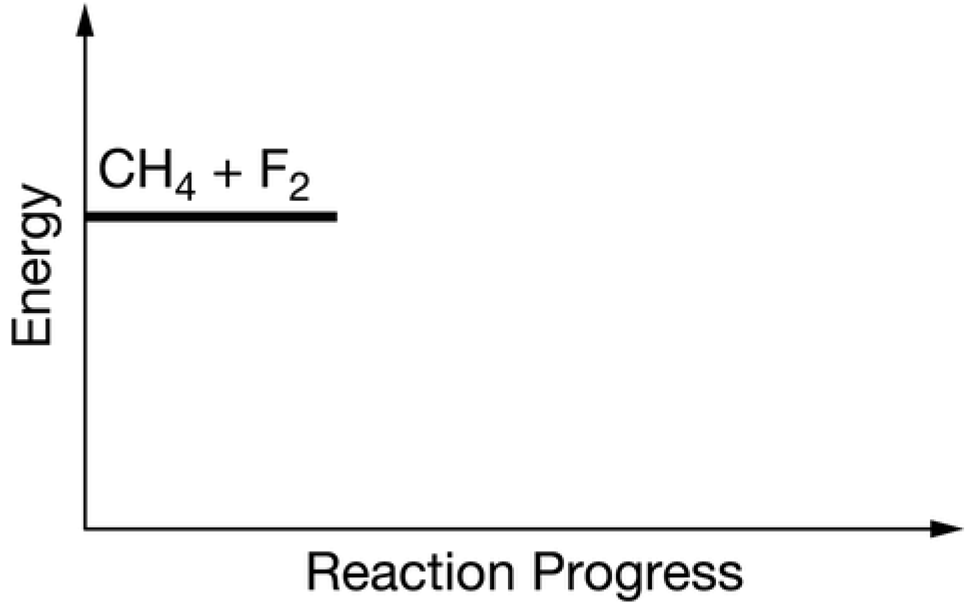
Please respond on separate paper, following directions from your teacher.

(c) The energy of the reactants is shown on the following energy diagram. On the right side of the energy diagram, draw a horizontal line segment to indicate the energy of the products. Draw a vertical double-headed arrow (↕) on the graph that corresponds to the value of ΔH for the reaction.



Please respond on separate paper, following directions from your teacher.

Part (a)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response meets all of the following criteria:

- ☐ Above the CH_4 is a gray circle representing a C atom attached to four black circles representing H atoms.
- ☐ Above the CH_3F is a gray circle representing a C atom attached to three black circles representing H atoms and one white circle representing an F atom.
- ☐ Above the HF is a black circle representing an H atom attached to a white circle representing an F

atom.

Part (b)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response gives the both the following set up and calculation:

$\Delta H = \Sigma$ enthalpies of bonds broken $- \Sigma$ enthalpies of bonds formed

$\Delta H = 413 + 155 - (485 + 567) = -484\ \mathrm{kJ/mol}_{rxn}$

(or $\Delta H = 4(413) + 155 - [3(413) + 485 + 567] = -484\ \mathrm{kJ/mol}_{rxn}$)

Part (c)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response. The response should be consistent with the response to part (b).

0	1	2 ✓

The response meets both of the following criteria:

- ☐ There is a horizontal line on the right side of the diagram that is lower than the horizontal line given on the left [but it should be higher to earn the point if a positive number was calculated in part (b)].
- ☐ There is a vertical double-headed arrow with one end at the level of the horizontal line given on the left and one end at the level of the horizontal line on the right.